

Anqi Wang

Marine Biogeochemistry & Ocean Carbon | Quantitative Environmental Science

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RESEARCH INTERESTS

Marine biogeochemistry & ocean carbon • Marine ecosystem modelling • Climate–ocean interactions
Data–model integration • Environmental machine learning • Ocean and Earth-system data

EDUCATION

MSc in Computational Methods in Ecology and Evolution | Merit

Imperial College London Oct 2022 – Oct 2023 United Kingdom

- Training in biological computing, Bayesian statistics, ecological modelling and data analysis in R/Python.
- Dissertation: *Modelling Seagrass Growth Patterns and Identifying Influencing Factors Using Cellular Automata for Restoration Purposes*. Developed a spatially explicit cellular-automata model to simulate five-year seagrass restoration trajectories under contrasting nutrient and current conditions.

BSc (Hons) in Environmental Science | Second Class Honours, Division One

University of Nottingham Sep 2019 – Aug 2022 United Kingdom

- Training in environmental microbiology, environmental pollution and quantitative modelling.
- Dissertation: *A General Model of Minimum Metal Concentration Co-Selection for Antimicrobial Resistance Optimised Using WHAM VII*. Used WHAM VII to model metal speciation and estimate environmental metal concentrations associated with antimicrobial-resistance co-selection.

RESEARCH EXPERIENCE

Research Assistant

Zhejiang A&F University, Jiyang College May 2025 – Present China

- Conduct quantitative research across hydrology, soil/agroecosystems and spatial ecology using multi-site observational, meteorological and geospatial datasets, contributing from data processing and modelling through interpretation, figure development and manuscript revision.
- Developed QC, event-detection and XGBoost/SHAP workflows to quantify non-rainfall water across diverse croplands, showing that its omission introduced a median **16.9% bias in water-input accounting**; the work led to a first-author manuscript in preparation and an oral presentation at WCSS 2026.
- Analysed **63 years of flood records across four major Yangtze stations** using nonstationarity diagnostics and GAMLSS, examining how flood distributions vary with hydroclimatic conditions and reservoir regulation.
- Integrated **24 years of multi-source data across 27 Yangtze River Delta cities** to examine ecological-resilience dynamics and carbon-related constraints using random forest modelling, SHAP-based interpretation and SEM; contributed quantitative analysis and interpretation to the manuscript under review at *Cities*.

Research Assistant

Shanghai Jiao Tong University Mar 2024 – May 2025 China

- Worked in an interdisciplinary collaboration between ocean/Earth-system science and electronic information engineering on machine-learning approaches for volcanic-signal detection and ice-core chronology assessment.
- Integrated ice-core chemical records with multiple volcanic-event chronologies and developed workflows for chronology alignment, signal-window feature engineering, model benchmarking and error analysis in noisy palaeoclimate records.
- Extended the analysis from event detection toward chronology assessment by using volcanic signals as candidate chronological tie points to evaluate age offsets and depth–age alignment against independently dated events; contributed to validation, scientific visualisation and manuscript development.

Research Collaboration – Marine Biogeochemistry

Tsinghua-affiliated Project Nov 2023 – Mar 2024 China

- Contributed to a marine biogeochemistry project investigating spatial variability in ocean carbon-cycle conditions and their physical and biogeochemical controls using global observational and Earth-system model datasets.
- Harmonised carbon-cycle, physical and biogeochemical variables across observational and model-derived datasets with different spatial and temporal structures, including consistency, coverage and quality checks.
- Constructed reproducible model-ready datasets for ensemble machine-learning analysis, cross-dataset comparison and scientific visualisation.

Research Intern

📅 Tsinghua University 📅 Mar 2022 – Jun 2022 📍 China

- Processed and integrated ocean-current time series and implemented R-vine copula models in R to characterise nonlinear multivariate dependence, covering data organisation, dependence diagnostics, model fitting and scientific visualisation.

Research Intern

📅 New York University 📅 May 2021 – Jul 2021 📍 Online

- Used simplified ocean-circulation and diffusion models to examine long-term pollutant transport and spatial spread in the Pacific Ocean; the project resulted in a conference paper.

PUBLICATIONS & MANUSCRIPTS

PUBLISHED

- **Wang, A.** (2023). The long-term diffusion of dumped wastewater from Japan in the Pacific Ocean. In *Environmental Pollution Governance and Ecological Remediation Technology* (pp. 261–269). Springer.

UNDER REVIEW / IN PREPARATION

- **Wang, A.**, et al. *From Shape to Signal: Machine Learning Detection of Volcanic Events in Ice Core Records*. In preparation.
- **Wang, A.**, et al. *Influence of Climatic Factors on Non-Rainfall Water Formation Across Diverse Croplands: Insights for Climate-Smart Agriculture*. In preparation.
- Chen, X., Huang, Z., **Wang, A.**, Xiao, Y., Zhang, T., Qin, H., & Guo, Z. *Nonstationary Flood Frequency Analysis and Drivers in the Middle-Lower Yangtze River*. Under review at *Journal of Hydrology: Regional Studies*.
- Fan, Y., Chen, X., **Wang, A.**, Zhou, S., He, H., Shi, L., Xiao, R., Miao, S., Qin, H., Guo, Z., & Zhang, X. *Spatiotemporal Evolution of Ecological Resilience, Catch-up Dynamics, and Carbon-carrying Constraints in the Yangtze River Delta Urban Agglomeration (2000–2023)*. Under review at *Cities*.

CONFERENCE PRESENTATION

23rd World Congress of Soil Science, Nanjing, China (Jun 2026) — Oral presentation: *Influence of Climatic Factors on Non-Rainfall Water Formation Across Diverse Croplands*.

TEACHING EXPERIENCE

Undergraduate Teaching

📅 Zhejiang A&F University, Jiyang College 📅 2025 – Present 📍 China

- Taught *Professional English for Environmental Engineering* and botany laboratory sessions, including lesson delivery, assessment preparation and coursework marking.
- Held office hours, supported undergraduate thesis projects, participated in final-year defences and assisted with course assessment and examination duties.

TECHNICAL SKILLS

Programming & scientific computing: Python, R, MATLAB, SQL, C, LaTeX; NumPy, pandas, scikit-learn, xarray/NetCDF.

Machine learning & statistical modelling: XGBoost, random forests, neural networks, GNNs, Transformers, SHAP, GAMLSS, Bayesian statistics, time-series analysis, feature engineering and model evaluation.

Ocean, environmental & geospatial data: observational and gridded datasets, Earth-system model outputs, multi-source data integration and quality control; ArcGIS, QGIS and Google Earth Engine.

Environmental modelling: ecological and spatial modelling, cellular automata, multivariate dependence modelling, data-model integration and scientific visualisation.

LANGUAGES

Mandarin Chinese (Native) • English (Fluent; IELTS 7.5) • Japanese (JLPT N2) • German (A2)